



How to find life on Venus?

Researchers suggest looking for spores in the atmosphere for signs of life on Venus. <http://digit.in/sep20-75>



NASA delayed due to COVID

NASA has delayed and revised its astronaut selection timeline due to the coronavirus pandemic. <http://digit.in/sep20-76>

GRAVITATIONAL WAVE ASTRONOMY

Understanding the universe through ripples in spacetime

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We do not know if the story of the apple falling on Newton's head is really true or not, but what we do know is that Newton noticed an object that was not flying away from the Earth – the Moon. Newton reasoned that unless there was some force that was pulling the Moon closer to the Earth, the motion of the Moon would cause it to move away in a straight line. Newton was the first to refer to this force as gravity, and described them using a simple equation that everyone learns in high school, known as the law of gravity. There were some aspects about the law that bewildered even Newton himself. For example, he admitted that he could never explain how the Earth exerts any kind of force on the Moon, across the vacuum of space. He understood that the force of gravitation applied universally between objects, or point masses, no matter what the distance between them, but could not explain how this was so. Newton also did not figure out the gravitational constant, and the equation assumed its final form once Henry Cavendish did that. The problem was, no one imagined at

that time that gravity propagated through space in the form of waves, and instead believed that the effect of gravity was instantaneous. Newton was however able to explain the observations of Kepler and provide evidence that the Earth went around the Sun. Two astronomers, Urbain Le Verrier and JC Adams independently predicted the existence of Neptune because the predicted orbits of the planets did not match the observations. The discovery of Neptune in 1846 based on the calculations by Urbain Le Verrier proved that Newton had been right about gravity.

James Clerk Maxwell showed that electricity, magnetism and light were all intertwined, and electromagnetic waves can propagate through space in the form of light. Maxwell himself wondered if gravity could also travel through fields in the form of waves, but did not follow up this thought. Oliver Heaviside, an English electrical engineer noted the similarity of the inverse square law in gravitation and electricity, and worked out a form of Maxwell's equation for gravity. Heaviside anticipated the existence of gravitational waves, and proposed that they travelled at a finite velocity,

instead of instantaneously. Albert Einstein then developed a theory of gravity from scratch, saying it was caused by warping of the spacetime continuum. His general theory of relativity gave rise to all kinds of preposterous concepts, such as regions of spacetime around which light could bend, or time itself behaving differently for clocks moving at different speeds. Over the years many of these theories would be proved. Einstein's theory had another implication, all accelerating objects create ripples in space time, that propagate through the continuum itself. This can be as simple as a swing, see-saw or merry go round in a playground, but the biggest gravitational waves were caused by exotic and then theoretical objects such as black holes. However, even for the strongest objects that Einstein could think of he never himself believed that they would ever be strong enough to be directly detectable. Newton found his own concept of gravity absurd, and Einstein remained skeptical about gravitational waves throughout his lifetime. According to General Relativity, the gravitational waves have a finite speed, the same as that of light.